

CO1 AT01

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?(5 Marks)

```
# Probability of selecting each section
P_A = 1/4
P_B = 1/2
P_C = 1/4

# Probability of selecting a blue car from each section
P_Blue_A = 3/10
P_Blue_B = 6/15
P_Blue_C = 2/15

# Total probability of selecting a blue car
P_Blue = (P_Blue_A * P_A) + (P_Blue_B * P_B) + (P_Blue_C * P_C)

# Bayes' Theorem
P_C_given_Blue = (P_Blue_C * P_C) / P_Blue

print("Probability that the blue car came from Section C:")
print(P_C_given_Blue)
print("Percentage =", round(P_C_given_Blue * 100, 2), "%")
```

```
Probability that the blue car came from Section C:
0.1081081081081081
Percentage = 10.81 %
```

Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

1. (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
3. (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
5. (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
6. (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
7. (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit. ii) Suggest two effective techniques to reduce overfitting in this model. iii) Discuss how overfitting affects the model's generalization ability on unseen data.

```

import matplotlib.pyplot as plt

# Data
epochs = [1, 2, 3, 4, 5, 6, 7, 8]
training_loss = [2.8, 2.5, 2.2, 2.0, 1.8, 1.6, 1.5, 1.3]
validation_loss = [2.4, 2.2, 2.0, 2.1, 2.3, 2.6, 2.9, 3.2]

# Plot
plt.figure(figsize=(8,5))
plt.plot(epochs, training_loss, marker='o', label='Training Loss')
plt.plot(epochs, validation_loss, marker='s', label='Validation Loss')

# Highlight overfitting point
plt.axvline(x=4, linestyle='--', label='Overfitting Starts')

plt.title("Training Loss vs Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.xticks(epochs)
plt.grid(True)
plt.legend()

plt.show()

```



3.Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks) 12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

19. (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%

i) Identify the epoch at which the model starts overfitting. ii) Explain why validation accuracy decreases despite training accuracy increasing. 22. iii) Suggest two methods to improve validation performance.

```
import matplotlib.pyplot as plt

# Data
epochs = [1, 2, 3, 4, 5, 6, 7, 8]
training_accuracy = [60, 65, 70, 75, 80, 85, 88, 92]
validation_accuracy = [58, 63, 68, 72, 73, 72, 70, 68]

# Plot
plt.figure(figsize=(8,5))
plt.plot(epochs, training_accuracy, marker='o', label='Training Accuracy')
plt.plot(epochs, validation_accuracy, marker='s', label='Validation Accuracy')

# Highlight overfitting point
plt.axvline(x=6, linestyle='--', label='Overfitting Starts')

plt.title("Training Accuracy vs Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy (%)")
plt.xticks(epochs)
plt.grid(True)
plt.legend()

plt.show()
```

